METRIC

MIL-PRF-32662
13 August 2020

PERFORMANCE SPECIFICATION

ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

1. SCOPE

1.1 Scope. This specification covers high-loading rate adhesives that are purposed for structural and armor applications. Examples include composite armor systems and other up-armoring or applique' systems.

1.2 Classification. Adhesives bonding systems are classified based on room temperature single-lap-joint performance under the following groups.

1.2.1 Group I. Group I adhesives are categorized as having a maximum strength (S_{max}) greater than or equal to 10.0 MPa (1450 psi) and a minimum complete failure displacement limit ($d_{failure}$) greater than 3.81 mm (0.15 in); (e.g., $d_{failure} > 3.81$ mm [0.15 in] and $S_{max} \geq 10.0$ MPa [1450 psi]).

1.2.2 Group II. Group II adhesives are categorized as also having a maximum strength greater than or equal to 10.0 MPa (1450 psi), yet their complete failure displacement limits fall between the range of greater than or equal to 1.60 mm (0.063 in) and less than or equal to 3.81 mm (0.15 in); (e.g., 1.60 mm [0.063 in] $\leq d_{failure} \leq$ 3.81 mm [0.15 in] and $S_{max} \geq 10.0$ MPa [1450 psi]).

Comments, suggestions, or questions on this document should be addressed to Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Specifications & Standards Office, Attn: FCDD-RLW-MC, Aberdeen Proving Ground, MD 21005-5069. Since contact information may change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 8040

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1.2.3 Group III. Group III adhesives are categorized with a maximum strength greater than or equal to 10.0 MPa (1450 psi) but producing complete failures at a displacement less than 1.60 mm (0.063 in); (e.g., $d_{failure} < 1.60$ mm [0.063 in], $S_{max} \geq 10.0$ MPa [1450 psi]).

1.2.4 Group IV. Group IV adhesives are categorized as having a maximum strength less than 10.0 MPa (1450 psi) regardless of the amount of displacement at complete failure.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-48611	-	Adhesive System, Epoxy-Elastomeric, for Glass to Metal.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-3059	-	Acceptance Criteria for Adhesives for High-Loading Rate Applications

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ARMY REGULATIONS

Army Regulation (AR) 25-400-2 - The Army Records Information System (ARIMS)

(Copies of these documents are available online at:

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https://dmna.ny.gov/milpay/forms/AR_25_400_2.pdf.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASQ/ANSI

ASQ/ANSI Z1.4-2003 (R2018)	Sampling Procedures and Tables for Inspection by Attributes

(Copies of these documents are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D1002	-	Standard Test Method for Apparent Shear Strength of Single-Lap-Joint Adhesively Bonded Metal Specimens by Tension Loading (Metal-to-Metal).
ASTM D1151	-	Standard Practice for Effect of Moisture and Temperature on Adhesive Bonds
ASTM D5229/D5229M	-	Standard Test Method for Moisture Absorption Properties and Equilibrium Conditioning of Polymer Matrix Composite Materials
ASTM E865	-	Standard Specification for Structural Film Adhesives for Honeycomb Sandwich Panels

(Copies of these documents are available online at https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION Publications

ISO/IEC 2382:2015(en)	-	Information Technology – Vocabulary
ISO 2859-1:1999(E)	-	Sampling Procedures for Inspection by Attributes
ISO 13008	-	Information and Documentation -- Digital Records Conversion and Migration Process

(Copies of this document are available from https://www.iso.org/home.html.)

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SAE INTERNATIONAL STANDARDS

SAE AMS 3695	-	Adhesive Film, Epoxy Base, For High Durability Structural Adhesive
SAE AMS 4035	-	Aluminum Alloy Plate & Sheet (2024)
SAE J243	-	Methods of Tests for Automotive-Type Sealers, Adhesives, and Deadeners
SAE J1524	-	Method of Viscosity Test for Automotive Type Adhesives, Sealers, and Deadeners

(Copies of these documents are available online at https://www.sae.org/.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Approval. Adhesives qualified under this specification shall be products that have been tested, and passed the approval tests specified herein.

3.2 Materials.

3.2.1 Adhesive. The adhesive shall be a thermosetting resin.

3.3 Bond Strength (Groups I, II, III and IV). The bond strengths for the Groups I, II, III and IV adhesive bonding systems shall be as specified in table I, as determined per guidance provided in MIL-STD-3059. Average bond strengths shall be measured from a minimum of 5 specimens for each testing condition.

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TABLE I. Bond strength, failure extension, and crack extension requirements. [1/]

Tier I Sample Conditioning	Tier I Test Temperature	Adhesive Classification	Minimum Avg. Strength	Minimum Avg. Failure Displacement
Ambient room temperature, dry	23° ± 2°C (73 °F, +/–3 °F)	Group I	≥ 10.0 MPa (1450 psi)	> 3.81 mm (0.15 in)
		Group II	≥ 10.0 MPa (1450 psi)	≤ 3.81 mm (0.15 in) ≥ 1.60 mm (0.063 in)
		Group III	≥ 10.0 MPa (1450 psi)	< 1.60 mm (0.063 in)
		Group IV	< 10.0 MPa (1450 psi)	Not Applicable
Tier I Single-Lap-Joint strength and failure displacement requirements, Section 4.5.4.1				
Tier II Sample Conditioning	**Tier II Test Temperature**	**Adhesive Classification**	**Minimum Avg. Strength Retention**	
Water immersion for 14 days at a temperature of 63 °C,+/–3 °C (145 °F, +/–5 °F)	23° ± 2°C (73 °F, +/–3 °F)	Group I	75% of initial dry maximum strength measured in first tier testing, no requirements for failure displacement	
		Group II		
		Group III		
		Group IV		
Tier II Single-Lap-Joint strength and failure displacement requirements, Section 4.5.4.2				
Tier II – Part 2 Sample Conditioning	**Tier II- Part 2 Test Temp.**	**Adhesive Classification**	**Minimum Avg. Strength Retention**	
Ambient room temperature, dry	71 °C, +/–3°C (160 °F, +/–5 °F)	Group I	50% of initial dry maximum strength measured in first tier testing, no requirements for failure displacement	
		Group II		
		Group III		
		Group IV		
Tier II – Part 2 Single-Lap-Joint strength and failure displacement requirements, Section 4.5.4.3				
Tier III Sample and Test Conditioning		**Adhesive Classification**	**Minimum Avg. G_{ISCC}**	
60 °C, +/–1.1 °C, (140 °F, +/–2 °F) and 100% relative humidity (RH) for 5 weeks		Group I	0.61 KJ/m^2 (3.5 in-lb/in^2)	
		Group II		
		Group III		
		Group IV	Not Applicable	
Tier III Crack Extension Test requirements, Section 4.5.1.4				
G_{ISCC} - Mode I tensile strain energy release rate while undergoing stress corrosion cracking.				

[1/] Averages shall be measured from a minimum of 5 specimens for each testing condition.

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3.4 Storage life. The Group I, Group II, Group III, and Group IV adhesive bonding systems shall meet the bond strength requirements of tests for Tier I, II, and III of table I, if applicable, after storage for 6 months at a temperature not to exceed 30° ± 2°C in the absence of sunlight. Storage stability shall also be maintained at an elevated temperature of 40° ± 2°C for two weeks (14 days).

3.5 Product characteristics.

3.5.1 Adhesive component.

3.5.1.1 Viscosity. 80 to 2500 Pa-s as measured with a rotational (Brookfield®) method. The adhesive should be hand trowelable or extrudable from cartridge containers using a hand-operated or pneumatic static mixing gun.

3.5.1.2 Pot life/Open time. For manual application, the open time life of the adhesive when mixed for preparing a serviceable bond at 23° ± 3°C, shall be a minimum of 15 minutes.

3.5.1.3 Sag and Bridging. The adhesive will have thixotropic static flow characteristics.

3.5.1.4 Curing pressure. The adhesive shall be capable of curing and making satisfactory bonded joints as described within this specification when only contact pressure is applied during the period of cure.

3.5.1.5 Curing time and temperature. The adhesive should harden to a workable state within a time frame of 15 minutes to 3 hours (at room or elevated temperatures) and should reach full bonding strength within 7 days at room temperature.

3.6 Instruction sheet. The manufacturer shall provide a dated, coded and titled instruction sheet with each shipment of adhesive outlining instruction for its use. The instruction sheet shall include the following information:

a. Complete recommended pre-bonding treatments and cleaning processes.
b. Maximum useable storage and pot life of the adhesive. The pot life of the adhesive after the addition of the activator (for multi-component adhesives) shall be based on definite quantities of the adhesive.
c. Mixing instructions including recommended percentage of activator and any temperature control necessary when mixing the adhesive.
d. Application instructions, including application temperature and relative humidity, and acceptable thickness range.
e. Typical cure conditions for the adhesive.
f. Necessary related data and precautions to be observed throughout all operations, including:

    1. Standard operating procedure (SOP) in the use of the material.
    2. Physical data on the characteristics of the components (boiling point and flash point).
    3. Fire and explosion hazard data including guidance on the proper protection to use if a fire occurs in the area.

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4. Reactivity data - guidance on classes of material which are not compatible with the components in the adhesive.
5. Precautions to take in the event of a spill or leak.

g. Any other pertinent information relative to the use and storage of the adhesive and or primer.

4. VERIFICATION

4.1 General provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection and testing of the adhesive shall be classified as follows:

a. Approval tests (see 4.5)
b. Quality conformance tests (see 4.6)

4.3 Test conditions.

4.3.1 Specimen conditions. All specimens shall be conditioned at 23° ± 2°C and for at least 24 hours. Other conditioning shall be prescribed in table 1. Preparation of the test specimens shall be in accordance with 4.4 and as described herein.

4.3.2 Normal conditions (room temperature). Single-lap-joint strength properties shall be determined for room temperature adhesive at 23° ± 2°C, no sooner than 10 minutes after specimens have reached equilibrium at such temperature.

4.3.3 Hot/Wet conditions. Conditioning for hot/wet strength retention tests shall require submersion of lap-joint specimens (i.e., non-ambient moisture conditioning) in a water immersion tank for fourteen (14) days at a constant temperature of 63 °C +/–3 °C (145 °F +/–5 °F). At the completion of the conditioning, specimens will be pat-dried and tested no later than thirty (30) minutes after being removed from the water immersion tank.

4.3.4 Elevated temperature conditions. Conditioning of single-lap-joint specimens for elevated temperature shear strength retention tests shall require a suitable oven to maintain the specimen at 71 °C, +/–3°C (160 °F, +/–5 °F) for a period of 45 minutes prior to, and during, testing.

4.3.5 Humidity conditions. Crack extension test samples will be exposed to elevated temperature and moisture conditions of 60 °C, +/–1.1 °C (140 °F, +/–2 °F) and 100% relative humidity (RH) for 5 weeks.

4.4 Preparation of test specimens.

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4.4.1 Preparation of single-lap-joint test specimens. Specimens for testing all the table 1 properties shall be prepared from 2024-T3 aluminum conforming to ANSI/SAE AMS 4035. The single-lap-joint test specimens are to be assembled per ASTM D1002, shown in figure 1. The plate thickness shall be 1.62 +/– 0.125 mm (0.064 +/– 0.005 in). Cleaning, surface pretreatment, and bonding procedures shall be in accordance the manufacturer's instruction sheet.

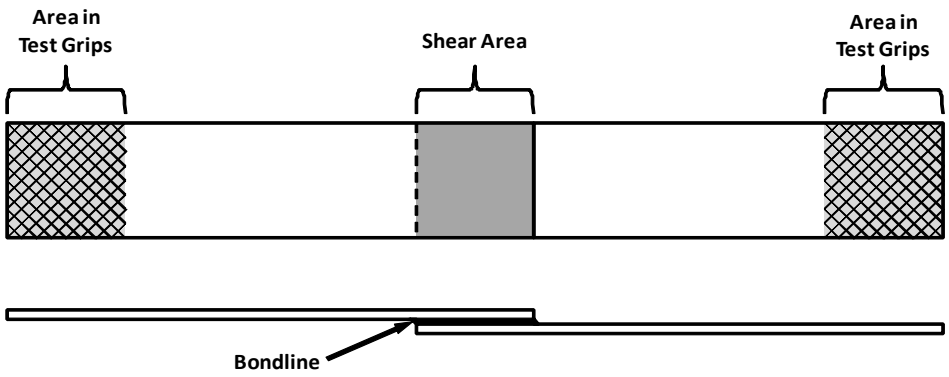


FIGURE 1. Assembly for adhesively bonded single-lap-joint test specimen configuration (refer to ASTM D1002 for dimensions).

4.4.2 Preparation of crack extension test specimens. Mode I tensile strain energy release rate while undergoing stress corrosion cracking (G_{ISCC}) will be performed, as described in SAE-AMS-3695. Double cantilever beam (DCB) samples shall be prepared by bonding two 300 mm x 350 mm 2024-T3 aluminum plates with a separator film (25 mm – 38 mm) along one edge, per SAE-AMS-3695, as shown in figure 3. Plate thicknesses shall be 12.7 mm. Samples shall be cut into widths of 25.4 mm. Tapped bolts will be used to initiate an opening displacement (y) of 2.54 mm. Figure 2 shows the DCB test specimen configuration.

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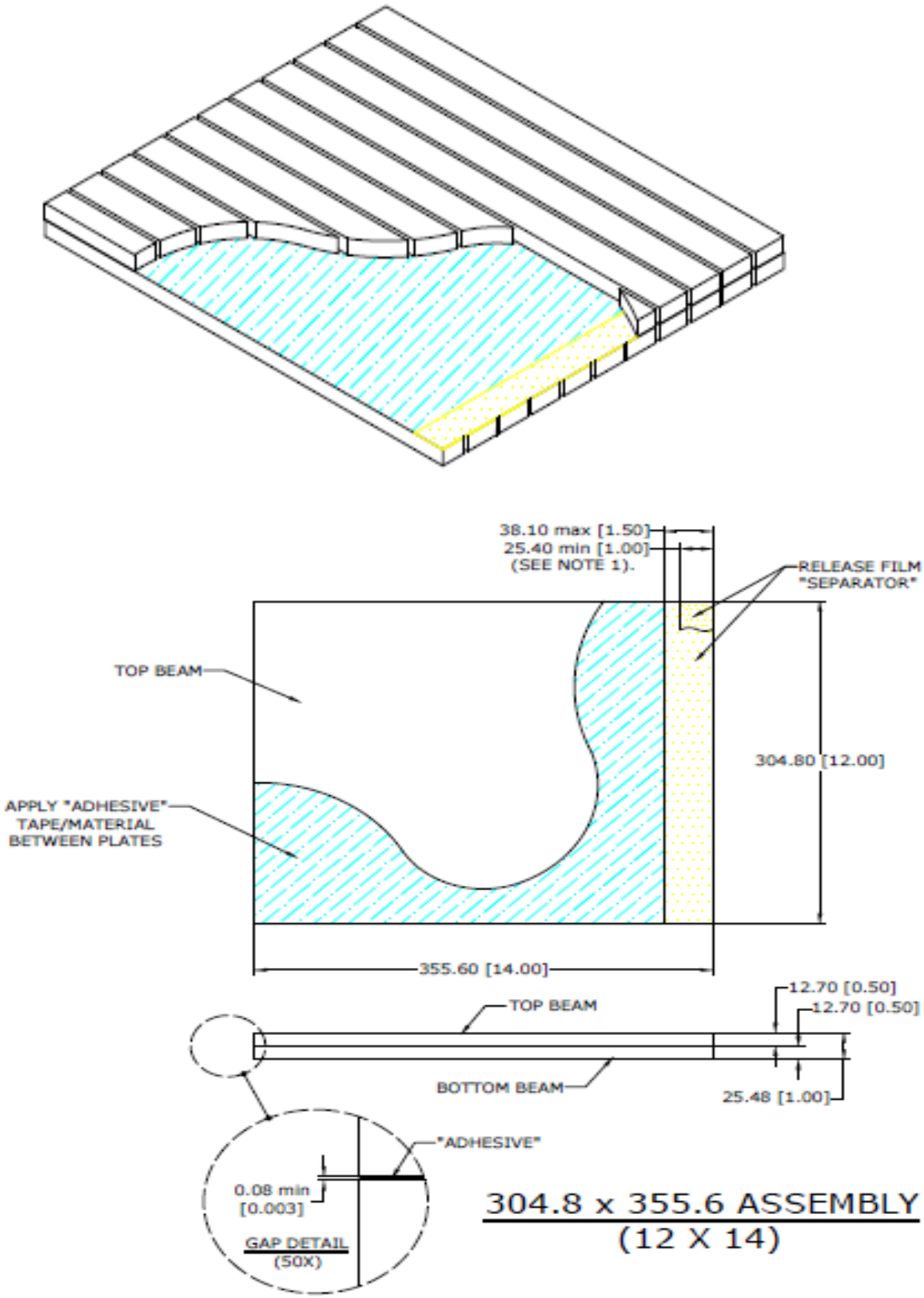


FIGURE 2. Bonded plate configuration as specified by SAE-AMS-3695.

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4.5 Approval tests.

4.5.1 Approval retest. Unless otherwise specified by the approving activity, adhesives which have previously passed the approval tests but have subsequently been modified in any manner shall satisfactorily pass re-testing of the material in order to retain approval. Approval tests may be repeated on material previously found satisfactory at any time at the option of the activity responsible for approval.

4.5.2 Manufacturer's data. Approval samples shall be accompanied by a test report showing results of all tests required by this specification.

4.5.3 Instruction sheet. Two copies of the manufacturer's instructions for use of the adhesive shall be furnished with the approval samples (see 3.6).

4.5.4 Tests. The approval tests shall consist of all the tests of this specification.

4.5.4.1 Tier I: Single-lap-joint strength and failure displacement test. (Test No. 1 of table I) The test specimen shall be gripped tightly and uniformly across the ends of the jaws of the test machine with the jaws and specimen so aligned that the jaws are directly opposite each other, and in such a position that an imaginary straight line will pass through the center of the bonded area and through the points of suspension. All specimens shall be pulled at a jaw separation rate of 1.3 mm per minute (0.05 in per minute). Tensile load is applied until the joint fails. The maximum effective strength (S_{max}) of the adhesive joint is defined as the maximum load (P_{max}) per shear area. Maximum effective strength shall be expressed in MPa of actual shear area, calculated to the nearest 0.65 cm^2. Displacement at failure ($d_{failure}$) is taken directly from the crosshead displacement of the testing machine and is expressed to the nearest 0.01 mm. A typical load versus displacement curve showing P_{max} and $d_{failure}$ is illustrated graphically in figure 3. A total of five specimens shall be tested. The average strengths and failure displacements of these five specimens shall be used to assign the adhesive Group classification.

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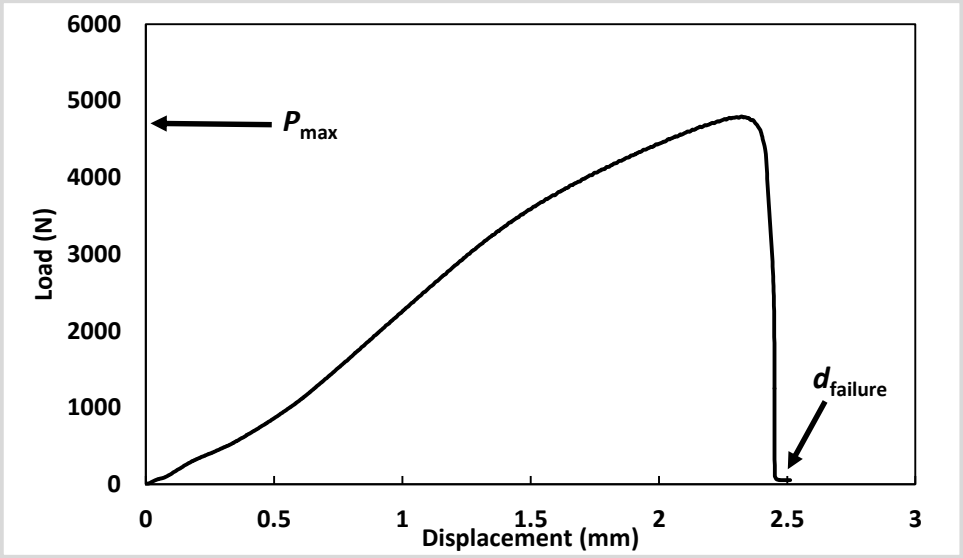


FIGURE 3. Graphical representation of P_{max} and $d_{failure}$ for a single-lap-joint tensile load versus displacement curve. $d_{failure}$ is defined as the displacement where the samples carries no load and the adherends have visibly separated.

4.5.4.2 Tier II: Single-lap-joint hot/wet strength retention test. (Test No. 2 of table I) The tensile shear test shall be in accordance with 4.5.4.1 after submersion of lap-joint specimens in a water immersion tank for fourteen (14) days at a constant temperature of 63 °C, +/–3 °C (145 °F, +/–5 °F). At the completion of the conditioning, specimens will be pat-dried and tested no later than thirty (30) minutes after being removed from the water immersion tank. A total of five specimens shall be tested. An average shear strength retention less than 75%, as specified in table I, shall cause rejection of the adhesive.

4.5.4.3 Tier II – Part 2: Single-lap-joint elevated temperature strength retention test. (Test No. 3 of table I) The 71° ± 3°C shear strength test shall be in accordance with 4.5.4.1 with the added requirement that a suitable oven shall be provided to maintain the specimens at the specified test temperature (See 4.3.4 and table I). Five specimens shall be tested for single-lap-joint test listed in table I. An average shear strength retention less than 75%, as specified in table I, shall cause rejection of the adhesive.

4.5.4.4 Tier III: Crack extension test. (Test No. 4 of table I) The crack extension test specimens shall conditioned in accordance with 4.5.3 at 60 °C, +/–1.1 °C (140 °F, +/–2 °F) and 100% relative humidity (RH) for 5 weeks. G_{ISCC} is determined at a constant loading tip displacement (y) by measuring the crack length (a) upon completion of 5 weeks of elevated temperature humidity exposure. G_{ISCC} can be found using the following equation.

$$G_{ISCC} = \frac{y^2 E h^3 [3(a + 0.6h)^2 + h^2]}{16[(a + 0.6h)^3 + ah^2]^2}$$

where:

y = opening displacement, 2.54 mm (0.100 in)

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E = Young's modulus, 73.08 GPa[1/] (10600 ksi) for 2024-T3 aluminum
h = height of one beam, 12.7 mm (0.50 in)
b = width of one beam (not shown), 25.4 mm (1.0 in)
a = crack length measured from the loading point

The expression for G_{ISCC} (KJ/m^2) can be simplified by substituting constant values for y, E, and h.

$$G_{ISCC} = \frac{0.00096578\left[3\left({}^{a}/_{1000} + 0.00762\right)^2 + 0.00016129\right]}{16\left[\left({}^{a}/_{1000} + 0.00762\right)^3 + \left({}^{a}/_{1000}\right)0.00016129\right]^2}$$

Crack length (a) is measured in millimeters.

The minimum acceptable value of G_{ISCC} after 5 weeks of exposure is 0.61 KJ/m^2 (3.5 in-lb/in^2). An average G_{ISCC} less than that specified in table I shall cause rejection of the adhesive.

4.5.4.5 Storage life tests. A 0.47 liter (0.50 US liquid quart) package of each component for adhesive shall be stored under temperature and time conditions specified in 3.4. At the end of the storage period, the components shall be used in the preparation of bonded lap joint specimens as required by test number 1 of table I. Five specimens shall be tested. An average shear strength retention less than 90% from initial testing prior to aging shall cause rejection of the adhesive.

4.5.4.6 Application life. Approximately 10 grams of the mixed adhesive shall be placed in a suitable container such as a plastic dish. A spatula with a 1 cm wide flat top shall be used in this test. Stir the adhesive frequently with the spatula and note the time at which the adhesive is considered to be definitely no longer spreadable on metal surface. The time to reach this consistency after mixing is the application life of the adhesive at 23° ± 3°C. The adhesive shall meet the requirements of 3.5.1.2 pot life/open time.

4.6 Quality conformance tests. Quality conformance tests shall consist of the sampling tests (4.6.1) and the inspection of filled containers (4.6.2).

4.6.1 Sampling for inspection and acceptance.

4.6.1.1 Adhesive. One container of adhesive shall be selected at random from each lot. The adhesive shall be subjected to the tests in 4.5.4.6. If the sample fails the application life test, the lot shall be rejected.

4.6.1.3 Lot. A lot shall consist of all the material (adhesive and curing agent/catalyst if applicable).

4.6.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected from each lot in accordance with inspection Level I, table II with Acceptance Quality

[1/] *ASM Handbook, Volume 2: Properties and Selection: Nonferrous Alloys and Special-Purpose Materials, ASM International, 1990.*

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Limits of 0.65 for major defects and 2.5 for minor defects. The sample containers shall be subjected to the inspection of filled containers as specified in 4.7.

4.7 Inspection of filled containers. Each sample filled containers selected in accordance with 4.6.2 shall be examined for defects of construction relative to the container and the closure, or evidence of leakage and for unsatisfactory markings. Each sample filled container shall be examined to determine the amount of contents. Any container in the sample having one or more defects or under required fill shall be rejected.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. (6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Group requirements are intended to show correlation against response measured in nonstandard Army testing configurations. For example, the groups defined in this Standard define property regions from bonded single-lap-joint tensile tests with performance in bonded armor configurations tested against ballistic threats. Grouping assignments/ categorizations of adhesives are intended to be amendable as Army needs change or are further refined. The grouping assignment/categorization is independent of the adhesive's chemical, processing, application, and property data and is based exclusively on single-lap-joint tensile data and performance. Multiple grouping assignments/categorizations for a given adhesive are possible under this convention. Therefore, an adhesive could potentially be assigned simultaneous high- and low-priority groups in differing Standards, depending on the chemical, processing, application, or property data used in the specific correlation. For example, adhesive requirements for armor and munitions have different properties. Therefore, evaluation places different weights on distinct performance characteristics. Grouping requirements/categorizations are specified using conventionally accepted testing standards, which are elaborated upon further within this Standard. The intention is to facilitate communication of U.S. Army–specific property requirements for research and development to industry, as well as academia.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Classification require (See 1.2).
c. Quantity desire

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d. Certified test report for this adhesive should be available upon request by the procuring activity.

e. Package marking should contain the following information:

1. Specification and type.
2. Manufacturer's designation.
3. Date of manufacture.
4. Recommended expiration date.
5. Storage requirements.
6. Precautionary markings.
7. Manufacturer's name and address.
8. Manufacturer's batch Identification.
9. Certified test report for this adhesive should be available upon request by the procuring activity.

f. A copy of the manufacturer's instructions should be enclosed with or attached to each container of adhesive as specified (see 3.6).

6.3 Drawings. For viewing ease and use with computer-aided design software, supplemental test panel and tooling fixture drawings (as PDFs and DXFs) are attached as Appendixes.

A. SINGLE-LAP-JOINT TEST PANEL DIMENSIONS
Drawing #1: Title: ASTM D1002 Optional Panel

B. SINGLE-LAP-JOINT TEST PANEL TOOLING FIXTURE DIMENSIONS
Drawing #2: Title: Single-Lap-Joint Tooling Fixture
Drawing #3: Title: Single-Lap-Joint Tooling Fixture – Top Plate
Drawing #4: Title: Single-Lap-Joint Tooling Fixture – Bottom Plate
Drawing #5: Title: Single-Lap-Joint Tooling Fixture – Dowel Pin
Drawing #6: Title: Single-Lap-Joint Tooling Fixture – Shim Plate

C. CRACK EXTENSION TEST PANEL AND DCB DIMENSIONS
Drawing #7: Title: Crack Extension Testing – SAE-AMS-3695 Assembly
Drawing #8: Title: Crack Extension Testing – SAE-AMS-3695 DCB Detail

D. SINGLE-LAP-JOINT FABRICATION CHECK LIST

E. SINGLE-LAP-JOINT MECHANICAL TESTING CHECK LIST

F. SINGLE-LAP-JOINT BONDLINE THICKNESS MEASUREMENTS

G. CRACK EXTENSION TEST SAMPLE FABRICATION CHECK LIST

H. CRACK EXTENSION TEST ENVIRONMENTAL CONDITIONING CHECK LIST

I. CRACK EXTENSION TEST BONDLINE THICKNESS MEASUREMENTS

J. CRACK EXTENSION TEST CRACK LENGTH MEASUREMENTS

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6.4 English units. When English divisions are required, units for meter, kilogram, meter per second, and mega Pascal may be converted to the English equivalent by multiplying them by the following conversion factors:

TABLE II. Metric SI to English unit conversion factors.

Metric SI unit	Divide by	Equals	English
meter (m)	0.0254	=	inch
meter (m)	0.3048	=	foot
kilogram (kg)	0.4534	=	pound
meter per second (m/s)	0.3048	=	feet/sec
mega Pascal (MPa)	0.006895	=	pounds/sq. inch

6.5 Definitions.

Database. As stated within ISO/IEC 2382-17:1999, a database should be “a collection of data organized according to a conceptual structure describing the characteristics of these data and relationships among their corresponding entities, supporting one or more application areas.”

Digital Archive. The term “digital archive” should be referred to as an electronic repository of digital data and metadata “to preserve the authenticity, reliability, integrity, and usability of such records” as stated in ISO 13008:2012(E).

Material Pedigree. The term “material pedigree” should refer to the documentation used to ensure the traceability of an adhesive and its constitutive formulation ingredients to the original manufacturer.

Metadata. Per ISO/IEC 2382-17:1999, the term “metadata” pertains to “data elements, possibly including their data descriptions, and data about data ownership, access paths, access rights and data volatility.”

Tier. The term “tier” should be considered a layer of testing protocols used to progressively characterize adhesive response. Each successive tier level requires an increasing commitment because of increasing experimental and analysis complexity.

6.6 Subject term (key word) listing.

Database
Material pedigree
Metadata
Single lap joint Testing

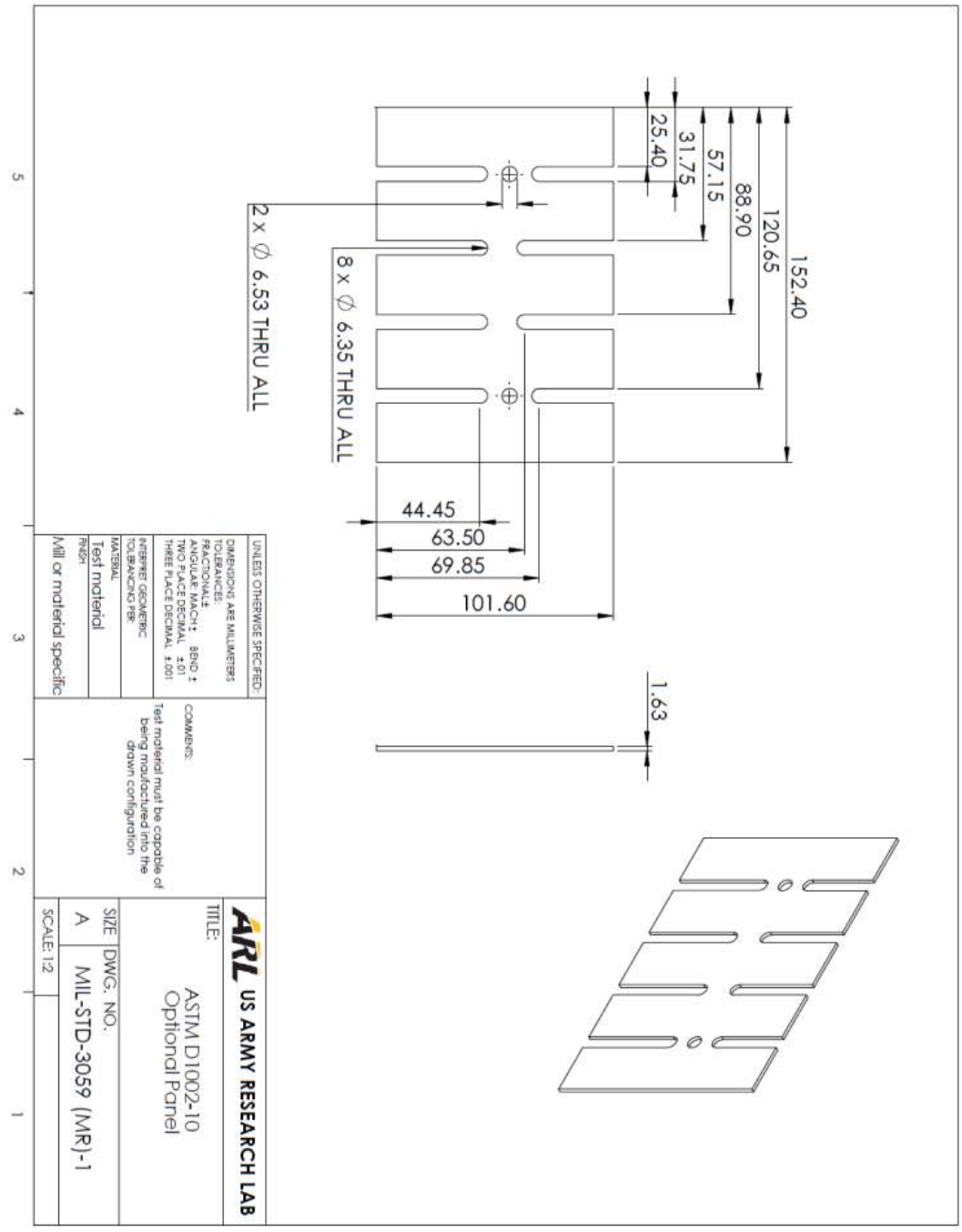
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SINGLE-LAP-JOINT TEST PANEL DIMENSIONS FOR ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

A.1 SCOPE

A.1.1 Scope. This appendix covers the ASTM D1002 single-lap-joint optional panel and its dimensions.

A.2 ASTM D1002 optional panel.



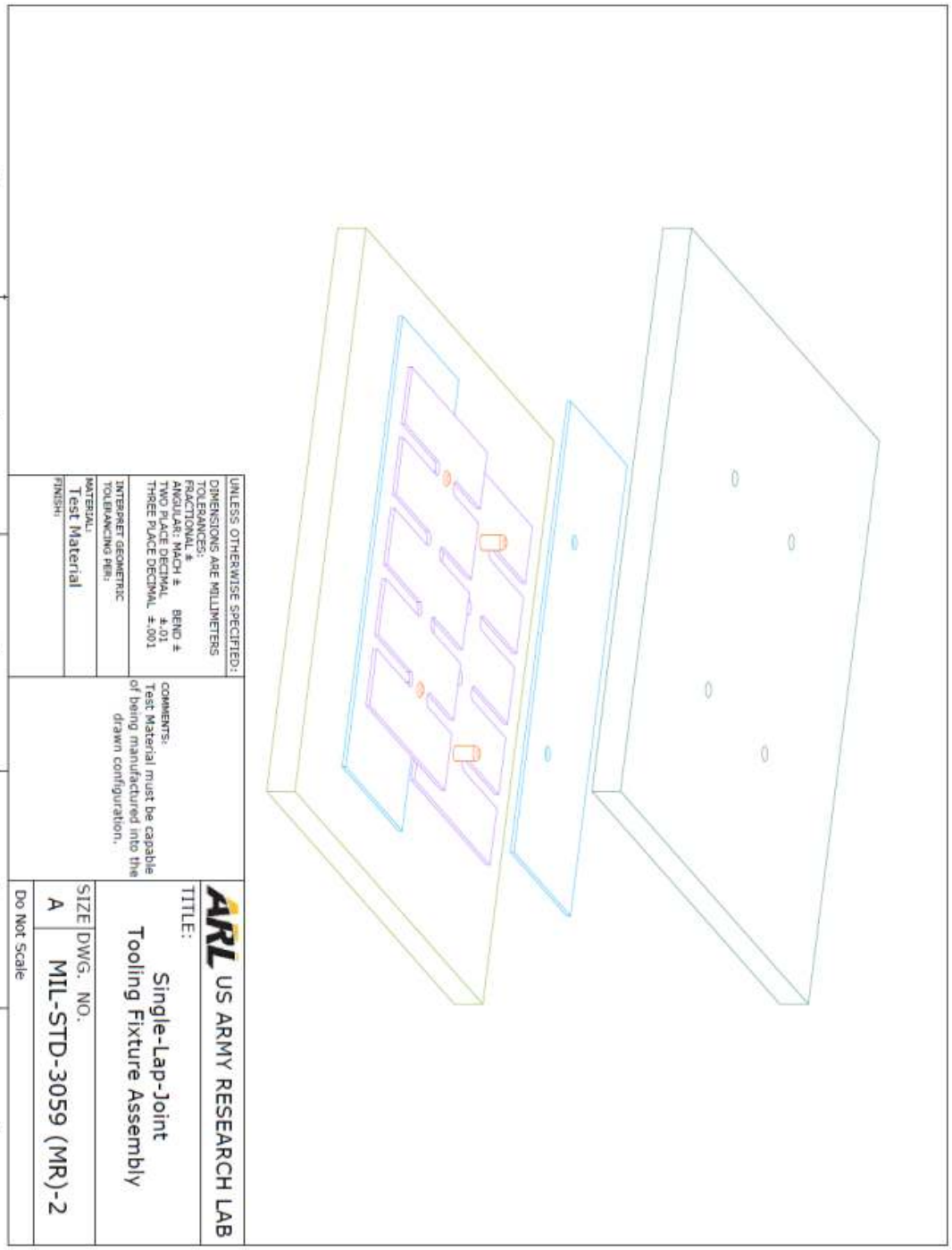
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SINGLE-LAP-JOINT TEST PANEL TOOLING FIXTURE DIMENSIONS FOR ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

B.1 SCOPE

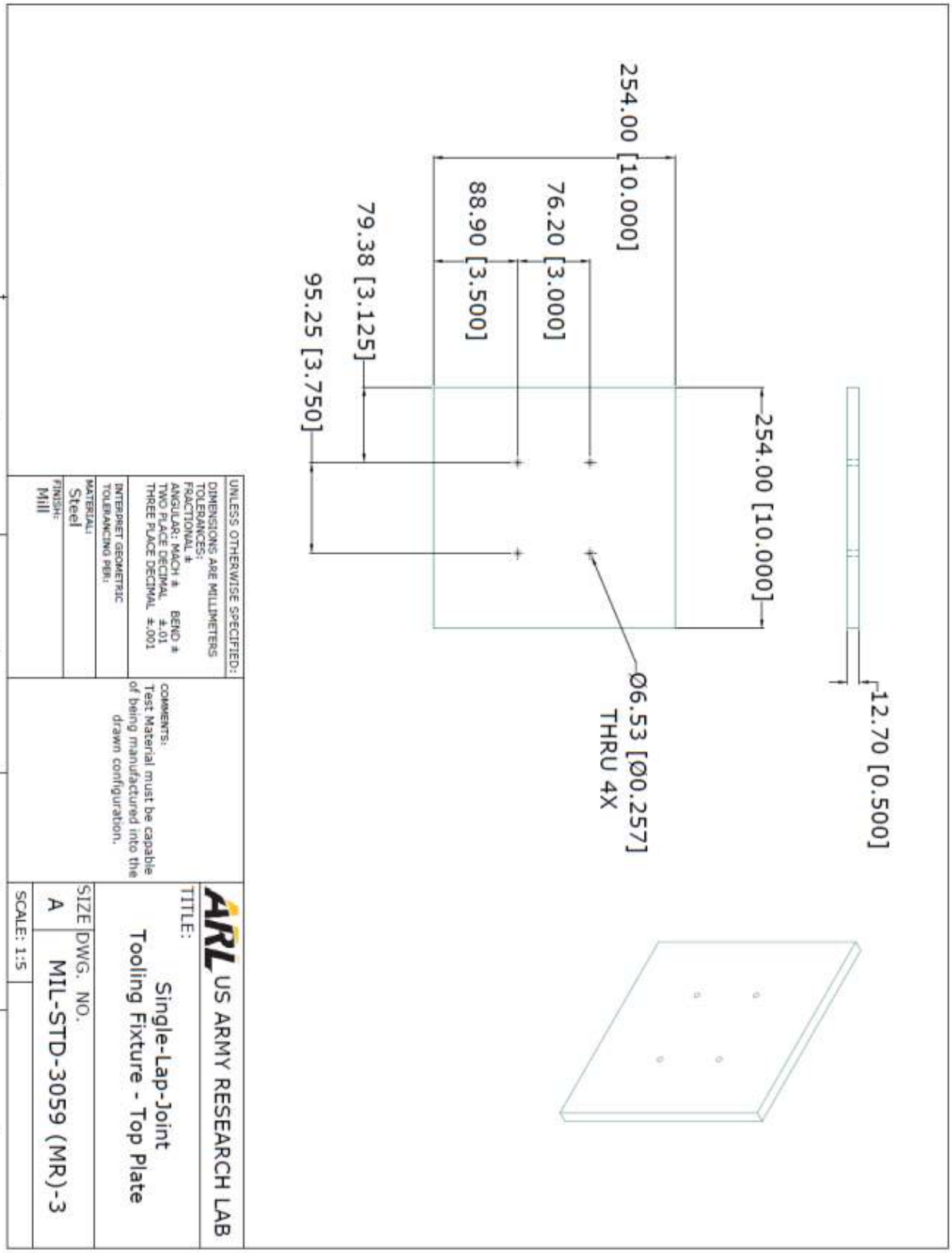
B.1.1 Scope. This appendix covers the single-lap-joint tooling fixture dimensions and its various components.

B.2 Single-lap-joint tooling fixture.



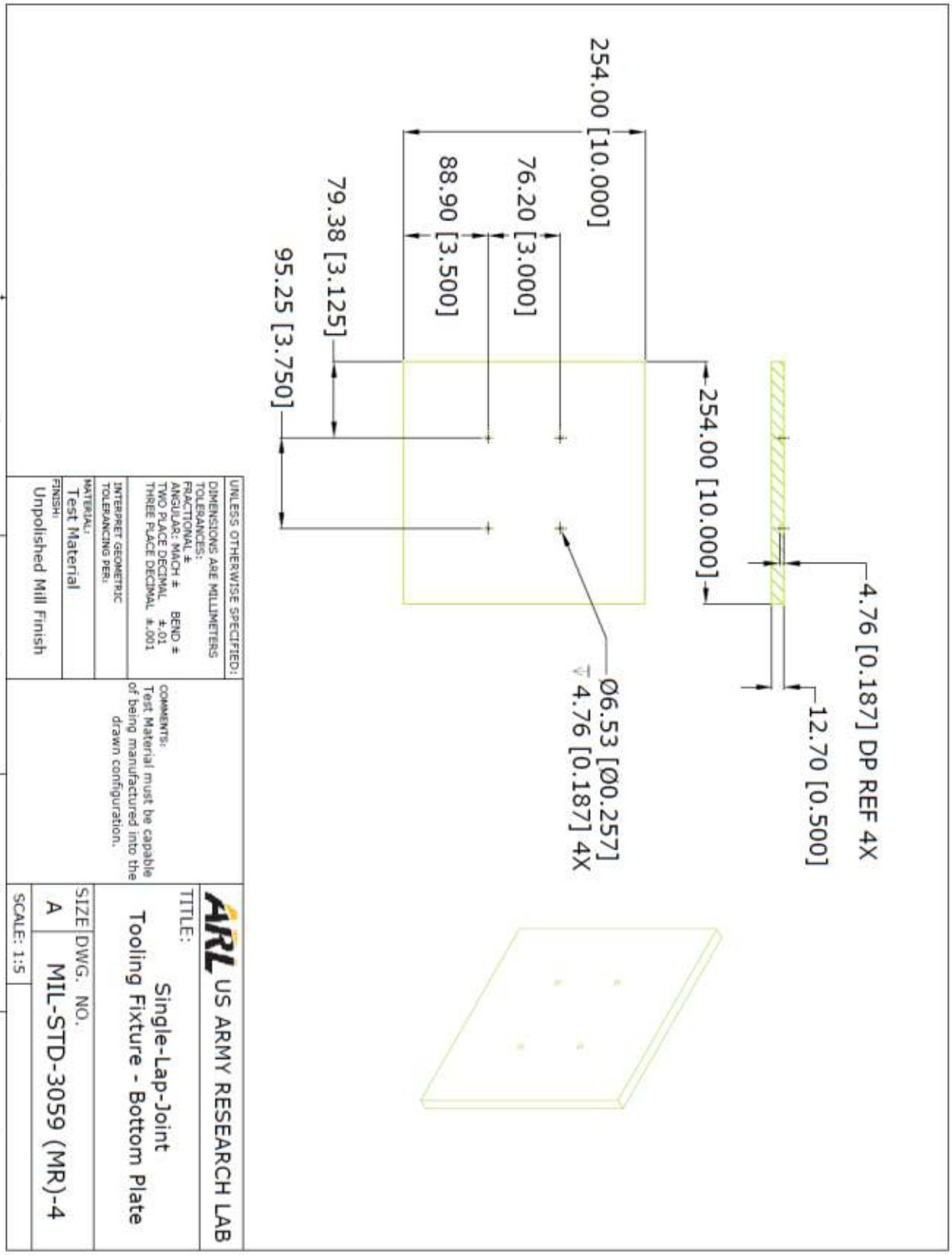
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B.3 Single-lap-joint tooling fixture – top plate.



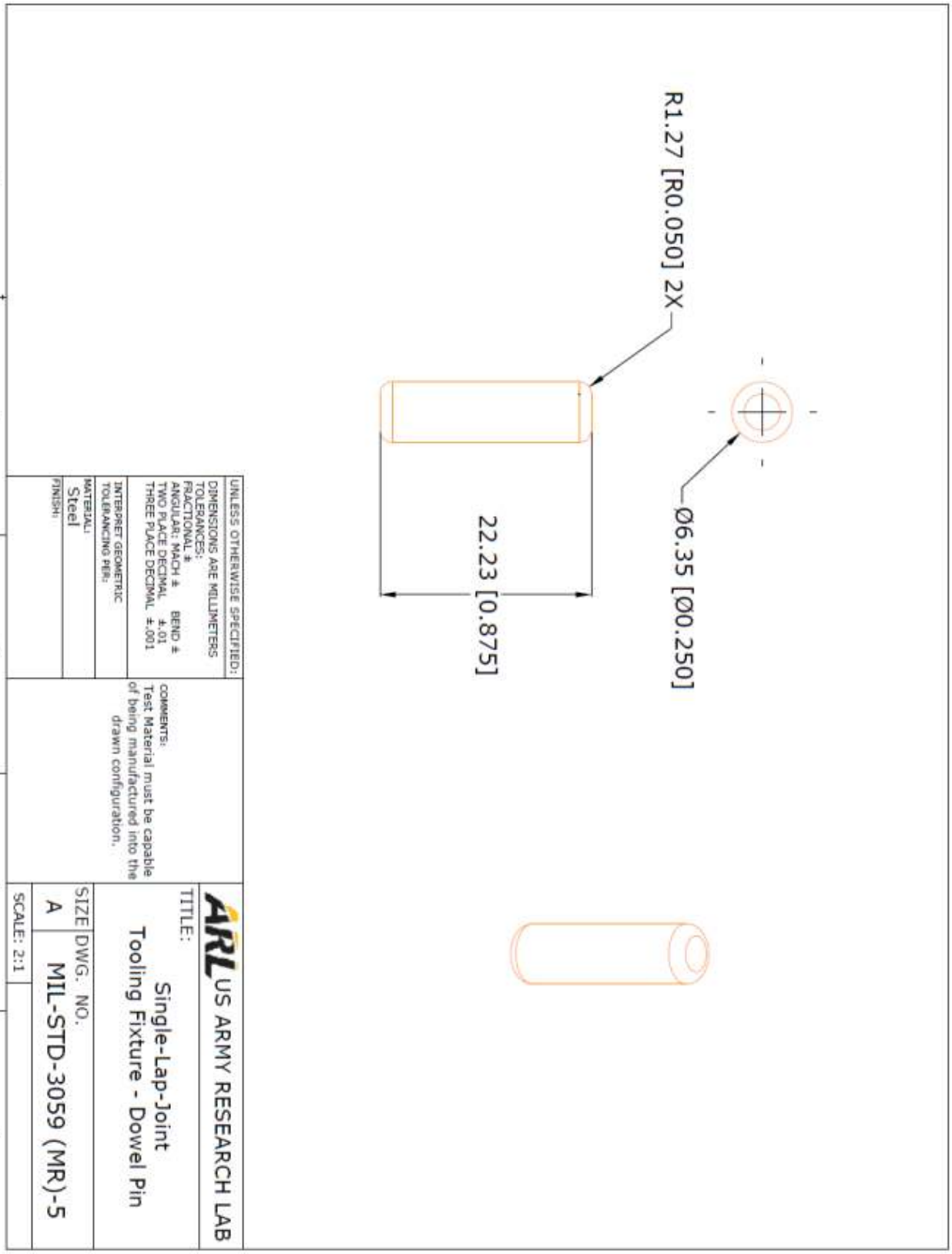
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B.4 Single-lap-joint tooling fixture – bottom plate.



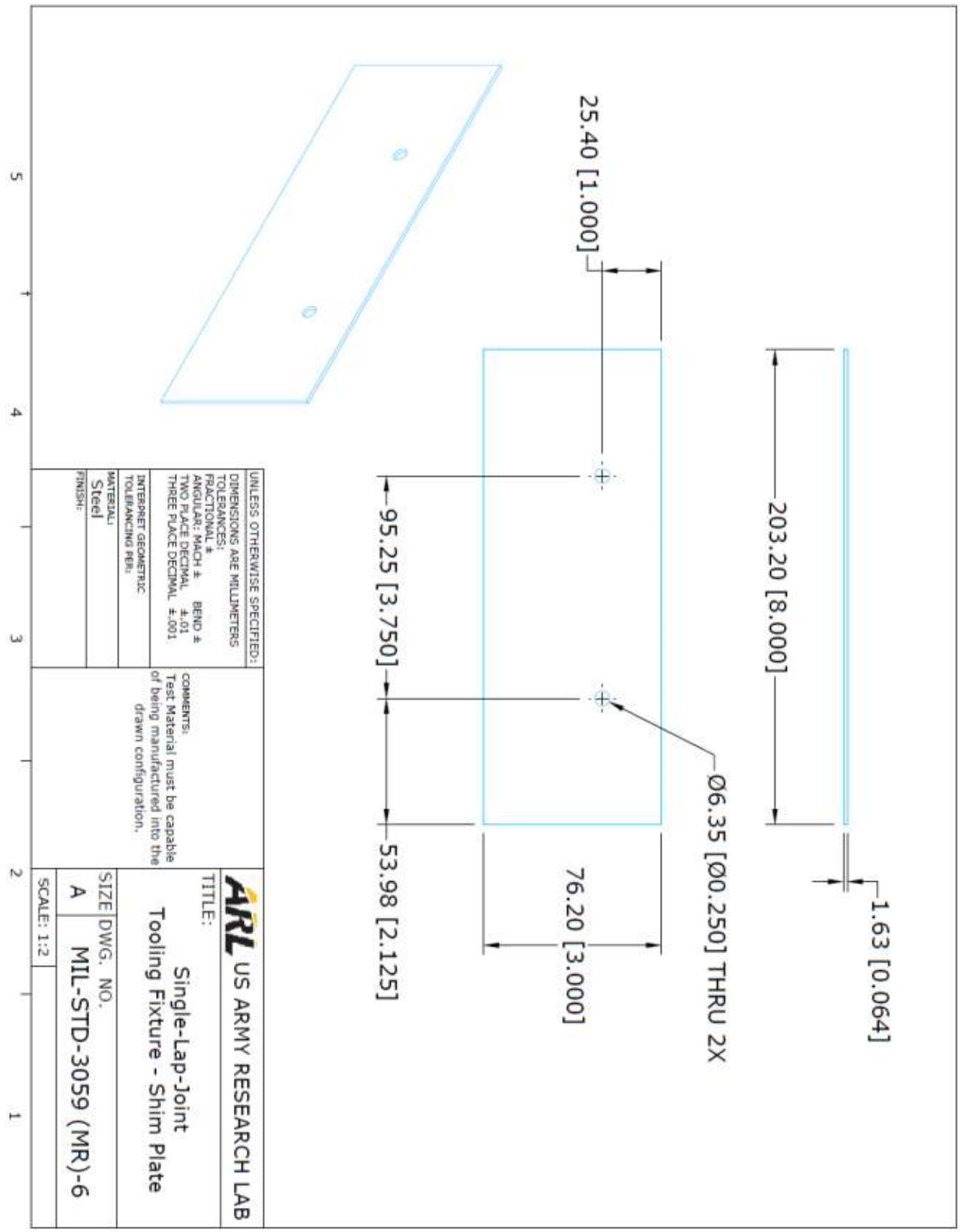
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:30Z
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B.5 Single-lap-joint tooling fixture – dowel pin.



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B.6 Single-lap-joint tooling fixture – shim plate.



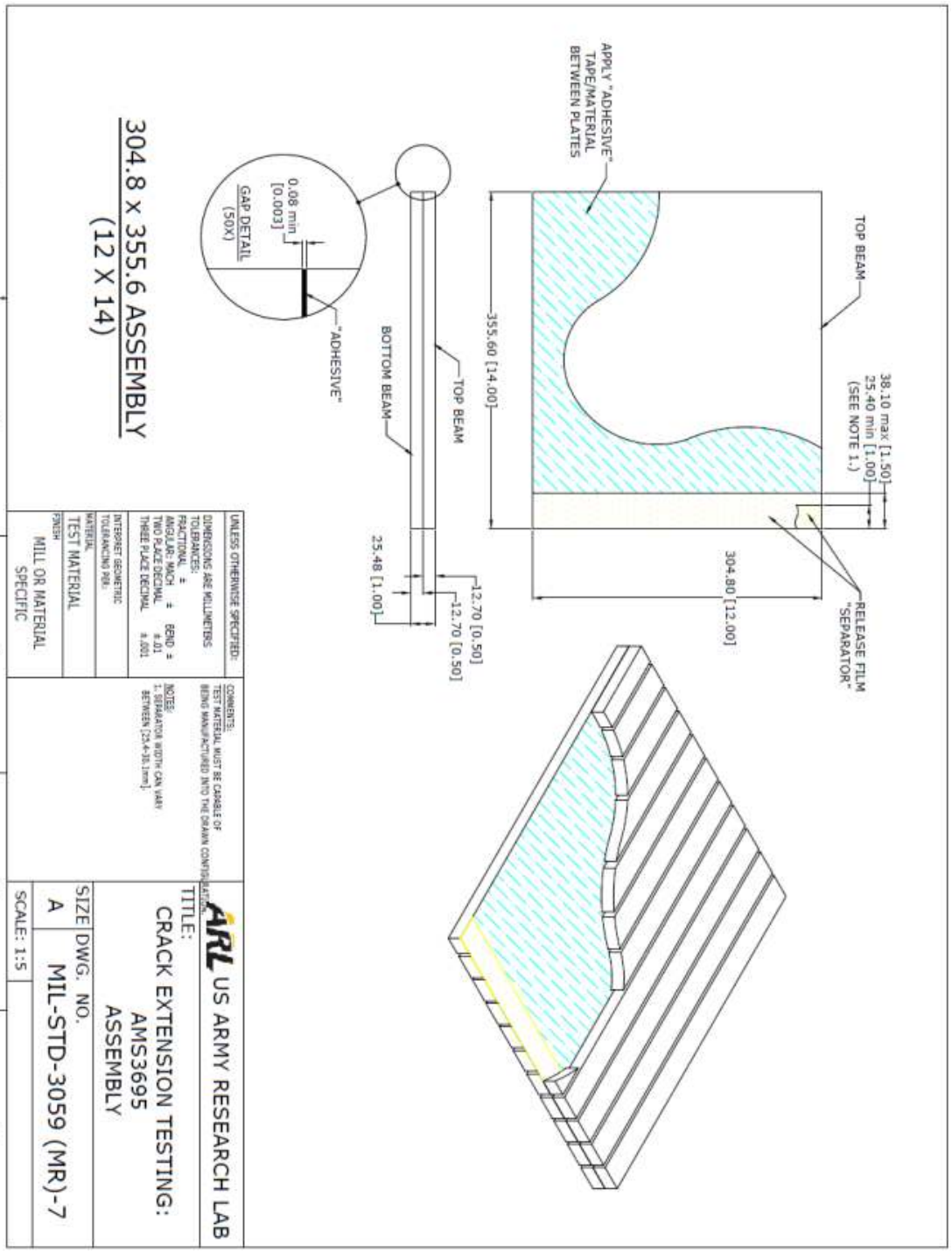
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CRACK EXTENSION TEST PANEL AND DCB DIMENSIONS FOR ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

C.1 SCOPE

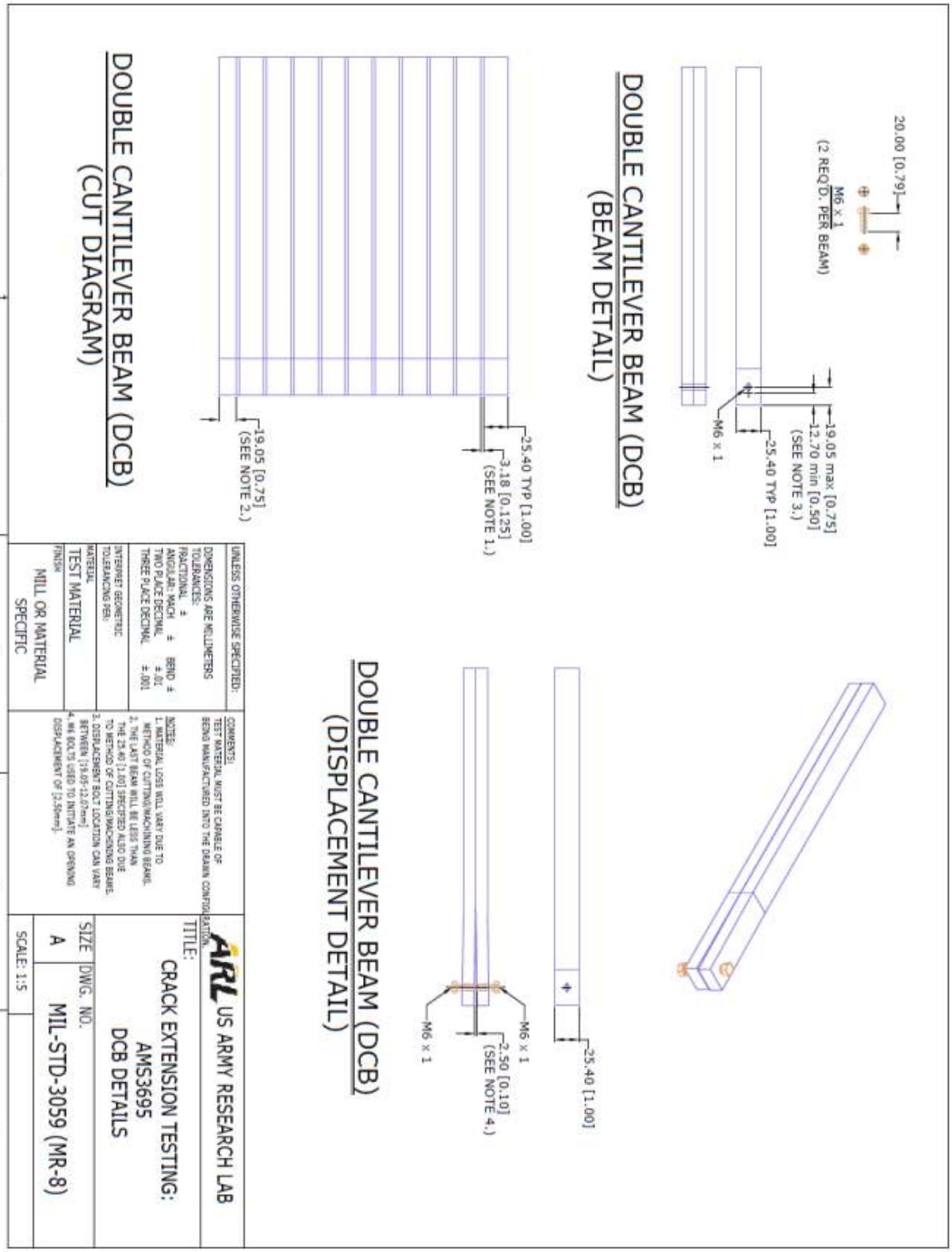
C.1.1 Scope. This appendix covers crack extension testing and the SAE-AMS-3695 assembly.

C.2 Crack Extension Testing – SAE-AMS-3695 Assembly.



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C.3 Crack Extension Testing – SAE-AMS-3695 DCB Detail.



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APPENDIX D

SINGLE-LAP-JOINT FABRICATION CHECK LIST FOR ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

D.1 SCOPE

D.1.1 Scope. This appendix covers the single-lap-joint fabrication check list.

Sample ID's:_____, _____, _____,_____,_____

Date: _____ Temperature:_____ Humidity_____

Testing facility and operator: _____

Adhesive, type, and form: _____

_____

Lot ID, manufacturer's code, expiration date: _____

_____

Method of cleaning and preparing bonding surface:_____

_____

_____

_____

_____

Adhesive preparation, application, and bonding conditions:

_____

_____

_____

Tooling fixture and bonding pressure notes: _____

_____

_____

Adhesive mixing to applied pressure time: _____

Cure cycle: _____

_____

_____

Oven used (model and serial number):_____

Conditioning procedure: _____

_____

_____

Comments:_____

_____

_____

_____

_____

_____

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SINGLE-LAP-JOINT MECHANICAL TESTING CHECK LIST FOR ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

E.1 SCOPE

E.1.1 Scope. This appendix covers the single-lap-joint mechanical testing check list.

Sample ID's:_____, _____, _____,_____,_____

Test date: _____ Test temperature:_____ Test humidity_____

Testing facility and operator: _____

Test frame model and serial number: _____

Test frame calibration date: _____

Load cell model and serial number: _____

Load cell calibration date: _____

Crosshead speed: 1.3 mm/min (0.05 in/min), list other_____

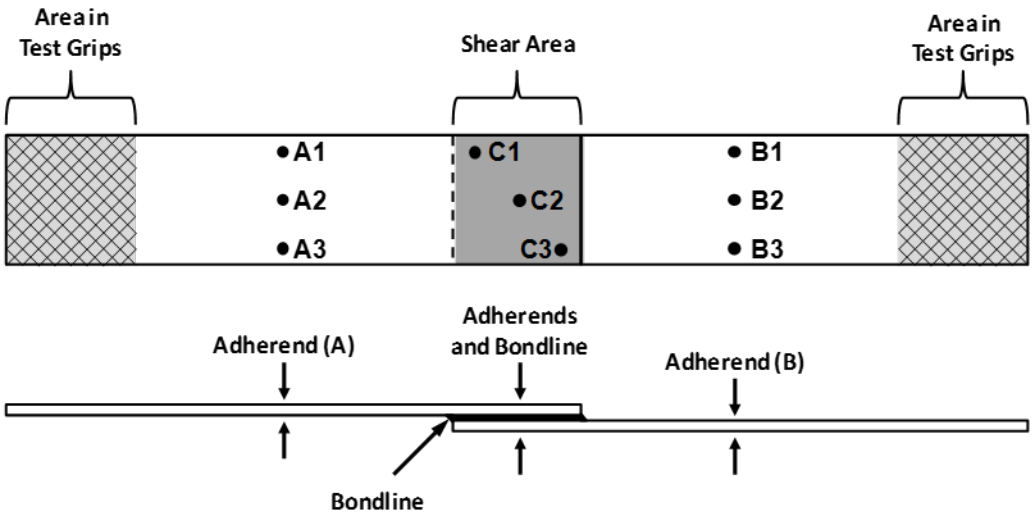
Nature of failure and other comments:

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SINGLE-LAP-JOINT BONDLINE THICKNESS MEASUREMENTS FOR ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

F.1 SCOPE

F.1.1 Scope. This appendix covers the single-lap-joint bondline thickness measurements.



Aluminum grade: _____ Mfg. nominal thickness:_____(mm – in)

Micrometer model, serial number, calibration date _____

Measurement units (circle) – mm or in

Sample ID	**Adherend (A1)**	**Adherend (A2)**	**Adherend (A3)**	**Average (A)**
Comments	**Adherend (B1)**	**Adherend (B2)**	**Adherend (B3)**	**Average (B)**
	Adherends and Bondline (C1)	**Adherends and Bondline (C2)**	**Adherends and Bondline (C3)**	**Average (C)**
				C – (A + B)
		Average bondline thickness		

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CRACK EXTENSION TEST SAMPLE FABRICATION CHECK LIST FOR ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

G.1 SCOPE

G.1.1 Scope. This appendix covers the crack extension test sample fabrication check list.

Sample ID's:_____,_____,_____,_____,_____,
_____, _____,_____,_____,_____,_____

Date: _____ **Temperature:**_____ **Humidity**_____

Testing facility and operator: _____

Adhesive, type, and form: _____
_____

Lot ID, manufacturer's code, expiration date: _____

_____

Method of cleaning and preparing bonding surface:_____
_____
_____
_____
_____

Adhesive preparation, application, and bonding conditions:

_____
_____
_____
_____

Plate alignment and bonding pressure notes: _____
_____
_____

Adhesive mixing to applied pressure time: _____

Cure cycle: _____
_____
_____

Oven used (model and serial number):_____

Conditioning procedure: _____
_____
_____

Comments:_____
_____
_____
_____
_____

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CRACK EXTENSION TEST ENVIRONMENTAL CONDITIONING CHECK LIST FOR ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

H.1 SCOPE

H.1.1 Scope. This appendix covers the crack extension test environmental conditioning check list.

Sample ID's:_____,_____,_____,_____,_____,
_____, _____,_____,_____,_____,_____

Test start date: _____ **Test temperature:** _____ **Test humidity**_____

Test end date: _____ **Test temperature:** _____ **Test humidity**_____

Testing facility and operator: _____

Environmental test chamber model and serial number: _____
_____

Environmental test chamber calibration date: _____

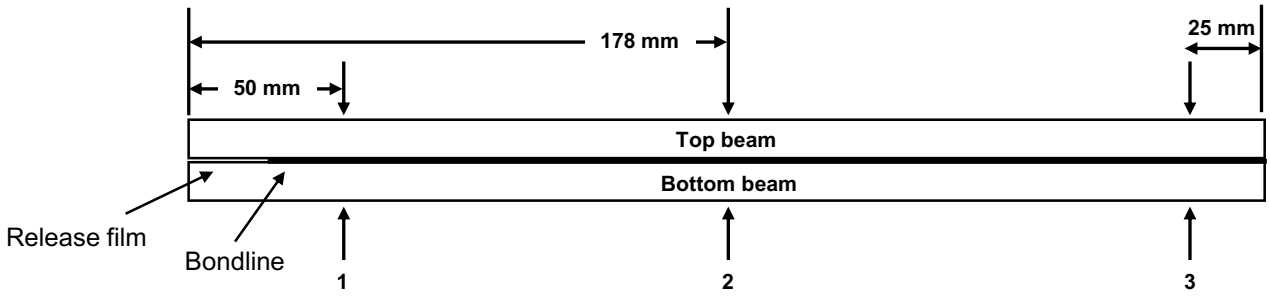
Nature of failure and other comments: _____

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CRACK EXTENSION TEST BONDLINE THICKNESS MEASUREMENTS FOR ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

I.1 SCOPE

I.1.1 Scope. This appendix covers the crack extension test bondline thickness measurements.



Aluminum grade:_____ **Mfg. nominal thickness:**_____**(mm – in)**

Micrometer model, serial number, calibration date _____

_____

Measurement units (circle) – mm or in

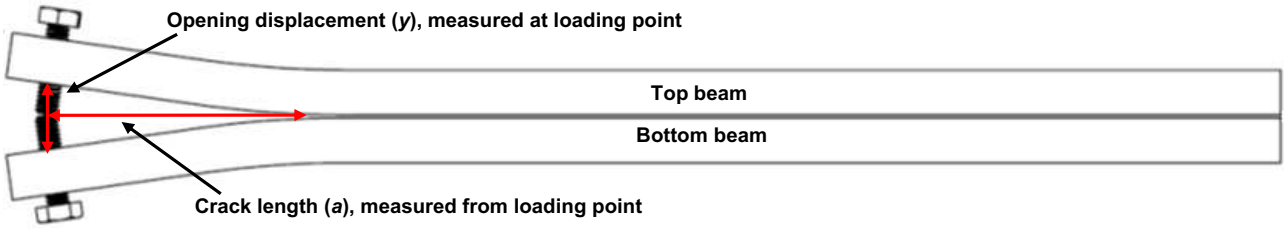
Sample ID	**Top beam height (A1)**	**Top beam height (A2)**	**Top beam height (A3)**	**Average (A)**
	Bottom beam height (B1)	**Bottom beam height (B2)**	**Bottom beam height (B3)**	**Average (B)**
	Beam and Bondline (C1)	**Beams and Bondline (C2)**	**Beams and Bondline (C3)**	**Average (C)**
				C – (A + B)
Comments		**Average bondline thickness**		

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CRACK EXTENSION TEST CRACK LENGTH MEASUREMENTS FOR ADHESIVE, HIGH-LOADING RATE, FOR STRUCTURAL AND ARMOR APPLICATIONS

J.1 SCOPE

J.1.1 Scope. This appendix covers the crack extension test crack length measurements.



Micrometer model, serial number, calibration date _____

_____

Measurement units (circle) – mm or in

Sample ID	**$a_{initial}$ (side 1)**	**a_{final} (side 1)**
Opening displacement (*y*)	**$a_{initial}$ (side 2)**	**a_{final} (side 2)**
Conditioning time (hours)	**$a_{initial}$ (average)**	**a_{final} (average)**
Mode-of failure (circle) – adhesive cohesive mixed-mode _____% adh. _____% coh.		
Comments:		

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CONCLUDING MATERIAL

Custodian:
Army - MR

Preparing activity:
Army – MR

(Project 8040-2020-001)

Review activities:
Army - AV, MI, PT
DLA- DH, IS, GS4

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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